

RESEARCH ARTICLE

Semantic Identifiability under Band-Order Ambiguity in PlanetScope Landslide Segmentation

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ABSTRACT

Released HR-GLDD NumPy arrays do not state their channel order. The paper’s Red–Green–Blue–NIR listing and notebook display support RGBN more strongly, but missing array-construction records leave native PlanetScope BGRN as a conservative sensitivity. For each contrast, we therefore report the range across both mappings and retain its direction only when that range excludes zero. Index-input-minus-raw and equal-mass-index-minus-plain F1 remain negative: $[-0.21, -0.01]$ and $[-0.41, -0.33]$ points. The boundary-F1 range $[-0.05, +0.38]$ crosses zero. Generic boundary weighting has a mapping-level mean range of $[+0.96, +1.13]$ points; all 18 seed/control effects are positive ($[+0.42, +1.62]$). CAS and LRD do not confirm transfer. A publicly locked Sen12Landslides test passes only the NDVI-modulated-minus-plain-boundary margin, $-0.06 [-0.41, +0.26]$ points; all arm means are 1.21–3.14% F1. Sen12 provides no support for recurrence and does not refute the mechanism. HR-GLDD remains descriptive because event and coordinate identities are absent.

KEYWORDS

landslide segmentation; PlanetScope; band-order ambiguity; schema robustness; boundary weighting; mechanism attribution

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1. Introduction

High-resolution optical imagery enables detailed landslide mapping, but roads, river deposits, clearings, and rock share similar structure. HR-GLDD provides 3 m PlanetScope patches (Meena et al. 2023); CAS, LRD, and Sen12 supply external multi-sensor or named-band tests (Xu et al. 2024, Orynbaikyzy et al. 2025, Höhn et al. 2025).

NDVI is attractive because fresh failures can disrupt vegetation, but roads, agriculture, riverbanks, and shadows also create index edges. An index-modulated boundary loss changes both boundary emphasis and spectral modulation unless ordinary boundary weighting is controlled. Array semantics are equally load-bearing: assigning Red to the wrong column changes both NDVI and raw-edge controls.

Devara et al. compare empirical and U-Net Sentinel-2 extraction (Devara, Maurya, and Dwivedi 2024), while Wang et al. design edge enhancement (Wang, Li, and Shi

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2025). Such work motivates boundary outcomes and generic controls before vegetation-specific interpretation.

Known-order channel permutations test adaptability (Li et al. 2025), not unresolved release provenance. We apply established identified-set logic to an evidentially asymmetric two-mapping audit with controlled reruns; we do not claim a new sensitivity primitive or equal support for the mappings.

We therefore evaluate components and semantic assumptions rather than presuming a mechanism. Our contributions are:

- (1) a candidate-mapping certificate that reports whether signed contrasts retain direction over the release-supported RGBN and conservative native-order BGRN mappings;
- (2) an executable audit that records RGBN’s stronger evidence, the absent array-construction provenance, and complete sensitivity runs;
- (3) capacity- and weight-matched zero, raw-edge, candidate-index, plain-boundary, and index-modulated controls;
- (4) outcomes selected without test-label threshold tuning, plus CAS, LRD, and publicly locked Sen12 stress analyses.

The certificate identifies negative index-versus-generic F1 directions and a positive generic boundary direction, but leaves the index-modulated boundary direction unidentified. None of three external analyses confirms transfer; their defects or degeneracy preclude robust rejection.

2. Related work

2.1. *Landslide segmentation*

U-Net and residual-block U-Net architectures provide standard segmentation baselines (Ronneberger, Fischer, and Brox 2015, Zhang, Liu, and Wang 2018). HR-GLDD is a PlanetScope benchmark (Meena et al. 2023), CAS provides multisensor RGB patches (Xu et al. 2024), and LRD and Sen12Landslides provide named Sentinel-2 data (Orynbaikyzy et al. 2025, Höhn et al. 2025). In Remote Sensing Letters, Devara et al. compare U-Net with an empirical Sentinel-2 rule (Devara, Maurya, and Dwivedi 2024). These performance studies do not isolate the active component.

2.2. *Spectral and edge guidance*

NDVI is a Red/NIR ratio, so a shared multiplicative term cancels algebraically; real mountain illumination still produces documented topographic sensitivity (Matsushita et al. 2007). TriGEFNet reports an NDVI-guided benefit (Zhang et al. 2026); spectral indices have also been combined with edge extraction (Zhang et al. 2023), and SAEENet adds generic boundary guidance (Wang, Li, and Shi 2025). Li et al. permute known HLS bands to test adaptability (Li et al. 2025); we instead audit unresolved released-array provenance with zero, raw, and equal-mass controls.

2.3. *Finite sensitivity and identified sets*

Functional ANOVA has shown that architecture, fine-tuning, learning strategy, and initialization contribute differently across remote-sensing classification datasets (Gho-

lami Shiri et al. 2026). That work decomposes variability across known configurations; here the uncertain input is physical channel identity. Multiverse analysis propagates reasonable coding choices (Steege et al. 2016), while partial identification retains sets under structural uncertainty (Manski, Sanstad, and DeCanio 2021). Qin and Tong make family-relative identified sets executable for evaluation claims (Qin and Tong 2026). Our min–max rule applies that established primitive to physical-schema uncertainty, with full retraining, same-class controls, and a hash-bound claim record.

2.4. Boundary weighting and evaluation

Pixel-wise boundary emphasis predates this application in U-Net (Ronneberger, Fischer, and Brox 2015). LSL-SS-Net uses a landslide-specific level-set loss, while E-MSLDNet reports edge-aware CAS segmentation with boundary metrics (Yang et al. 2024, Zhao et al. 2025). We do not propose another boundary architecture or loss; we test whether index-specific attribution survives plain-boundary same-class control and unresolved physical schema.

3. Controlled methods

3.1. Band semantics and spectral controls

Let $\mathbf{X} \in \mathbb{R}^{B \times 4 \times H \times W}$ denote the released HR-GLDD arrays. The paper lists Red, Green, Blue, and NIR, and the notebook directly displays columns 0–2 as RGB; RGBN therefore has stronger release-internal support. Neither source binds names to NumPy indices through array-construction or reordering code. We retain native-product BGRN as a lower-support conservative sensitivity, not an equally documented mapping. Green/NIR occupy columns 1/3 in both; Red is column 0 under RGBN and 2 under BGRN. For $o \in \{\text{RGBN}, \text{BGRN}\}$, define

$$V_o = \frac{X_{\text{NIR}} - X_{\text{Red}(o)}}{X_{\text{NIR}} + X_{\text{Red}(o)} + 10^{-6}}. \quad (1)$$

We compare three two-channel controls:

$$C_{\text{zero}} = (0, 0), \quad (2)$$

$$C_{\text{raw}, o} = (|\nabla X_{\text{Red}(o)}|, |\nabla X_{\text{NIR}}|), \quad (3)$$

$$C_{\text{index}, o} = (V_o, |\nabla V_o|). \quad (4)$$

The zero control preserves attention parameters while removing external spectral information. The raw control tests whether generic optical edges explain any candidate-index advantage. Every recorded HR-GLDD output declares its candidate order; neither candidate is declared authoritative.

3.2. Semantic-identifiability certificate

Let $\mathcal{O}$ be the finite audit set of declared candidate mappings and $\delta(o)$ a signed contrast under schema o . We define its semantic envelope

$$\mathcal{E}_\delta = \left[\min_{o \in \mathcal{O}} \delta(o), \max_{o \in \mathcal{O}} \delta(o) \right]. \quad (5)$$

The effect direction is semantically identified only when $0 \notin \mathcal{E}_\delta$; otherwise it is direction-unidentified. Here $\mathcal{O} = \{\text{RGBN}, \text{BGRN}\}$. This conservative sensitivity certificate neither assigns equal plausibility nor claims that BGRN is release-documented; it is not a confidence interval or an exhaustive schema set.

3.3. Capacity-matched residual attention

All controlled-attention models use the same residual encoder–decoder with channels (32, 64, 128, 256) and three skip gates. At scale l , the two control maps are bilinearly resized and concatenated with encoder features F_l :

$$M_l = \sigma[\text{Conv}_{1 \times 1}(\text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}[F_l, C_l])))], \quad (6)$$

$$F_{l,\text{out}} = F_l \odot M_l + F_l. \quad (7)$$

Every control mode therefore has 2,114,084 trainable parameters. U-Net and ResU-Net are included as architectural baselines.

3.4. Loss controls

For pixel i , the unreduced base-loss map combines binary cross-entropy and focal loss (Lin et al. 2017), with focal positive-class weight $\alpha = 0.35$ and focusing exponent 2:

$$\ell_{\text{base},i} = \frac{1}{2}\ell_{\text{BCE},i} + \frac{1}{2}\ell_{\text{focal},i}. \quad (8)$$

Let $B_{\text{GT}} = \text{Dilate}_{3 \times 3}(Y) - \text{Erode}_{3 \times 3}(Y)$ denote the two-sided, one-pixel-radius Chebyshev morphological boundary band. We compare:

$$L_{\text{plain}} = \text{mean}_i\{\ell_{\text{base},i}[1 + \gamma B_{\text{GT},i}]\}, \quad (9)$$

$$L_{\text{index},o} = \text{mean}_i\{\ell_{\text{base},i}[1 + \gamma \widetilde{B_{\text{GT}}|\nabla V_o|_i}]\}, \quad (10)$$

where $\gamma = 2$. For an informative index-boundary signal, the tilde rescales it within each image to the mean mass of B_{GT} ; when its mean is at most 10^{-6} , it falls back to B_{GT} . Under both candidate orders, 8 of 1,119 training images use this fallback and the maximum mass mismatch is 2.98×10^{-8} . Thus L_{plain} versus $L_{\text{index},o}$ changes spatial allocation without changing average added weight.

3.5. Evaluation quantities

We compute pooled and per-tile F1, intersection over union, precision, recall, FPR, and per-tile boundary F1 with one-pixel Chebyshev matching.

For operating-point comparison, every arm uses the exact validation-score break-point nearest U-Net’s threshold-0.5 validation recall; higher thresholds break ties. Test labels never select thresholds.

For same-sensor spatial degradation, PlanetScope test patches are antialiased $128 \rightarrow 38 \rightarrow 128$ before inference. This measures information loss, not Sentinel-2 radiometry.

4. Experimental design

4.1. Dataset and scope of inference

We use the released HR-GLDD arrays (Meena et al. 2023): 1,119 training, 284 validation, and 355 test patches of 128×128 pixels at 3 m ground sampling distance. Landslide pixels comprise 10.09%, 10.17%, and 10.92% of the three splits. The release contains imagery and masks but no event identifier, source-scene identifier, or coordinate for each array row. Separate patches therefore cannot be demonstrated to be independent. Results are descriptive across three training seeds; we do not calculate tile-bootstrap confidence intervals, p-values, spatial-block statistics, or event-held-out performance.

The official test set had also been inspected during earlier method development. The present experiment is consequently treated as corrective evaluation rather than untouched confirmation. It can compare controlled configurations on the released benchmark but cannot establish population-level geographic generalization.

4.2. Factorial controls

The residual-attention surface crosses zero, raw-edge, and candidate-index controls with base and plain boundary-weighted losses; one additional index arm uses equal-mass index-modulated weighting. U-Net and ResU-Net bring the total to nine configurations. We train all nine with three seeds under both RGBN and BGRN candidates (54 runs).

All arms use AdamW (Loshchilov and Hutter 2019) with learning rate 1.5×10^{-3} , weight decay 10^{-4} , cosine decay to 10^{-5} , batch size 32, and 15 epochs. Validation F1 selects the epoch. Seeds 42, 43, and 44 use identical schedules and deterministic data-loader seeds. Probabilities are stored as float32 arrays; an independent script recomputes the reported default-threshold metrics without importing the training or reporting functions.

4.3. Post-hoc CAS sensitivity

We compare base and plain-boundary loss for the same RGB zero-control model on designated CAS regions (Xu et al. 2024). The protocol was corrected after execution, so this is post-hoc sensitivity without confirmation credit; Supplementary Method S1 gives sampling and deviations.

4.4. Internally precommitted LRD stress test

LRD supplies named Sentinel-2 bands and 28 EIDs (Orynbaikyzy et al. 2025). A local 16/6/6 protocol preceded validation/protected access, but no third-party timestamp exists and PH0003 shares a trigger family with protected PH0001/PH0004. Supplementary Method S2 gives the frozen design and limitations.

4.5. Publicly locked Sen12 stress test

Sen12Landslides supplies named Sentinel-2 bands and inventories (Höhn et al. 2025). After development-only calibration, 21 checkpoints were frozen before authorized access to 1,889 patches from ten protected inventories. Five conditions covered two upper margins, generic-boundary recurrence, absolute non-degeneracy, and conservative transformed-envelope integrity. A public post-access correction treated one SCL-255 array as descriptively unavailable before inference; no input, mask, endpoint, or condition changed (Supplementary Method S3).

4.6. Interpretation rules

Input specificity requires the candidate index to outperform zero and raw controls on the same endpoint under both orders. Spectral loss specificity requires index modulation to beat equal-mass plain weighting and retain effect direction across orders; boundary F1 is separate. The certificate covers index-versus-raw F1, index-versus-plain F1 and boundary F1, and plain-boundary-versus-base F1. Complete-system improvement overrides neither control. Resolution attribution is restricted to same-sensor degradation. LRD is not confirmation-eligible; Sen12 recurrence requires all five registered conditions.

5. Results

5.1. Band-order sensitivity and matched controls

Table 1. Three-seed means under release-supported RGBN and conservative native-order BGRN sensitivity. BF1 uses one-pixel Chebyshev tolerance.

Configuration	RGBN F1	BGRN F1	RGBN BF1	BGRN BF1
U-Net/base	66.92	66.92	49.62	49.62
ResU-Net/base	68.55	68.55	52.93	52.93
Attn-zero/base	68.88	68.88	53.17	53.17
Attn-raw/base	69.05	69.13	52.92	53.15
Attn-index/base	69.04	68.92	52.76	52.83
Attn-zero/plain-boundary	69.99	69.99	56.04	56.04
Attn-raw/plain-boundary	69.88	70.19	55.85	56.15
Attn-index/plain-boundary	69.98	70.12	55.96	56.20
Attn-index/index-boundary	69.58	69.78	56.34	56.15

Table 1 reports every configuration under both candidate orders; Supplementary Table S4 gives detailed RGBN-candidate operating metrics. With base loss, candidate-index attention reaches 69.04% F1 under RGBN and 68.92% under BGRN. Raw candidate-Red/NIR edges reach 69.05% and 69.13%, while the capacity-matched zero

control reaches 68.88% under both. Candidate-index input specificity is therefore unsupported regardless of order. All contrasts and envelopes below use unrounded run-level values; subtracting the table’s two-decimal cells can differ by 0.01 point.

Plain boundary weighting raises F1 in every seed and control mode under both orders. Mean F1 spans 69.88–70.19% and boundary F1 55.85–56.20%. Equal-mass index modulation changes F1 by -0.41 points under RGBN and -0.33 under BGRN. Its boundary-F1 contrast changes from $+0.38$ to -0.05 points, so spectral modulation is not order-robust. The order-stable result is generic boundary emphasis, not spectral identity.

5.2. *Semantic-identifiability certificate*

Here, “certificate” means the range of a contrast over both mappings; it records whether the sign survives this finite sensitivity, not statistical uncertainty. The index-input-versus-raw F1 envelope is $[-0.21, -0.01]$ points and the index-versus-plain-loss F1 envelope $[-0.41, -0.33]$, identifying negative directions over the two declared candidates without assigning equal plausibility. The corresponding boundary-F1 envelope $[-0.05, +0.38]$ crosses zero and is direction-unidentified. Generic plain-boundary F1 has envelope $[+0.96, +1.13]$; all 18 individual effects lie between $+0.42$ and $+1.62$ points. Supplementary Table S5 is the generated certificate.

5.3. *External analyses*

The post-hoc CAS sensitivity fails three of four criteria: plain boundary weighting changes mean region-macro F1 by -2.19 points and boundary F1 by -1.86 points, with mixed seed effects. It receives no confirmation credit (Supplementary Table S1).

The internally precommitted LRD run fails all four criteria but is not confirmation-grade. Validation F1 is 0.008–0.151; selected-threshold mean F1 changes by -4.73 points. Its frozen boundary effect is -24.10 points, but alternatives span -0.15 to -4.92 , and two protected EIDs share a trigger family with development. It is failed/indeterminate, not a robust rejection (Supplementary Method S2).

The publicly locked Sen12 test is verified-not-confirmed. Its sole passing condition is the NDVI-modulated-minus-plain-boundary F1 margin: -0.06 points ($[-0.41, +0.26]$); the registered upper endpoint is below $+1.00$ point. NDVI-attention minus raw-edge F1 is $+0.70$ points ($[+0.28, +1.16]$) and fails its margin. Generic boundary F1 is $+0.61$ ($[-0.09, +1.32]$), with 5/10 inventories positive. All arm means are only 1.21–3.14% F1. The registered transformed-envelope check also fails for one Kyrgyzstan pair, although native rectangles only share an edge. This provides no support for recurrence and no refutation (Supplementary Method and Table S3).

5.4. *Operating point and spatial degradation*

At threshold 0.5, boundary-weighted residual-attention FPR is 2.47–2.50%. Validation-matched recall is exact in 53/54 runs; combined matched FPR/precision spans 1.97–2.01%/78.41–78.74%. This does not support index-specific false-positive suppression.

Controlled $128 \rightarrow 38 \rightarrow 128$ degradation lowers F1 by 4.33–5.20 points across orders, identifying same-sensor spatial-information sensitivity rather than cross-sensor causation.

6. Discussion

6.1. *What the controls change*

The semantic audit changes the evidentiary question. RGBN has stronger release-internal support, but absent array-construction provenance prevents treating its Red index as directly verified. BGRN is a conservative native-product sensitivity, not an equally evidenced mapping. Index-versus-generic F1 directions survive this audit set, whereas the index-modulated boundary envelope crosses zero.

Generic boundary weighting is more stable within HR-GLDD: all 18 seed/control/order comparisons improve F1. None of three external analyses confirms transfer. CAS is post hoc with mixed seed signs; LRD has weak validation, endpoint instability, and trigger-family leakage. Sen12 has the strongest prospective pedigree, but fails four of five registered conditions: absolute F1 is degenerate and the conservative transformed-envelope integrity bar fails. Native rectangles only abut, so this is not evidence of duplication. Together these analyses leave transfer unsupported, not robustly refuted.

This does not imply that NDVI is universally unhelpful. Devara et al. report useful spectral processing (Devara, Maurya, and Dwivedi 2024), and TriGEFNet reports an NDVI-guided multimodal benefit (Zhang et al. 2026). Our contribution is diagnostic: explicit semantic uncertainty and same-class controls, not competitive performance or confirmed transfer.

6.2. *Interpretation under incomplete observability*

Band-ratio cancellation is exact only for a shared multiplicative irradiance term. Mountain scenes can additionally contain diffuse illumination, cast shadow, adjacency, bidirectional reflectance, and atmospheric effects. An NDVI gradient also responds to any vegetation discontinuity. The experiment contains no illumination labels or background-type annotations; it cannot validate topographic invariance or isolate scarps from roads and riverbanks.

Varotsos and Zhou describe incompletely observable remote-sensing systems whose operational use requires dataset-specific calibration (Varotsos and Zhou 2026). We apply analogous caution by separating same-sensor degradation from cross-dataset shift and withholding event-level inference where group metadata are absent.

6.3. *Implications for benchmark practice*

Four practical recommendations follow. First, multispectral array releases should carry machine-readable channel names rather than relying on sensor conventions. Second, claims about geographic transfer or confidence intervals should be tied to event, scene, or spatial-block identifiers; a patch index is not an independence unit. Third, mechanism studies should include generic same-class controls. For spectral attention this means zero and raw-edge controls; for a specialized boundary loss it means ordinary boundary weighting with matched total mass. Fourth, unresolved schemas should be propagated into a machine-readable semantic envelope rather than collapsed to one convenient interpretation.

The strongest HR-GLDD configuration is compact, but the external sensitivity failure prevents presenting it as a generally useful boundary-aware model. Parameter count characterizes implementation cost only; it is not evidence of an end-to-end emer-

gency workflow.

7. Limitations

HR-GLDD does not bind band names to NumPy indices through array-construction provenance. RGBN has stronger paper/notebook support; BGRN is a conservative native-order sensitivity, not an equally supported schema. The finite certificate neither identifies wavelengths nor supplies statistical uncertainty. Missing event, scene, and coordinate mappings preclude clustered inference; prior test inspection makes evaluation corrective.

Binary masks omit background categories, so Supplementary Figure S1 supports visual inspection but not category-specific error rates. The 10-m control measures sensitivity to PlanetScope smoothing and resampling, not Sentinel-2 radiometry or geography.

The post-hoc CAS analysis jointly changes sensor, resolution, geography, and annotation. LRD has no independently timestamped pre-access record; two protected EIDs share a trigger family with development; 84.1% of protected crops are empty; and validation F1 is weak. Its frozen criteria fail, but the external conclusion is indeterminate.

Sen12 used development-driven calibration without confirmation credit; v6 corrected one descriptive SCL field after access but before inference. Arm F1 is below 3.2%. Its registered transformed-envelope bar fails for abutting Kyrgyzstan tiles despite zero native positive-area overlap, precluding clean confirmation or refutation.

Three seeds quantify optimization variation, not independent events. No regional mosaic, acquisition pipeline, geospatial reconstruction, hardware payload, or responder workflow was evaluated; operational deployment and broad geographic generalization remain outside scope.

8. Conclusion

HR-GLDD lacks array-construction provenance binding band names to NumPy indices: RGBN has stronger release support, while BGRN is a conservative native-order sensitivity. The resulting certificate gives negative index-versus-generic F1 envelopes, a positive generic-boundary envelope, and a zero-crossing index-modulated boundary envelope.

Ordinary boundary weighting improves every HR-GLDD seed/control/order comparison. None of three external analyses confirms transfer: CAS is post hoc, LRD is indeterminate, and the publicly locked Sen12 test satisfies only one of five conditions with degenerate absolute F1. Generic boundary weighting is therefore dataset-local evidence, not a transferable prior; Sen12 provides no support for recurrence but does not establish refutation.

Machine-readable semantics, schema-propagated certificates, same-class controls, independently timestamped protocols, and true independence units must precede physical or generalization interpretations.

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Data availability statement. The HR-GLDD arrays are at <https://doi.org/10.5281/zenodo.7189381>; LRD is at <https://doi.org/10.5281/zenodo.17007637>; and Sen12Landslides is at <https://huggingface.co/datasets/paulhoehn/Sen12Landslides>. Code, summaries, output hashes, and source/history assets are in <https://github.com/akssha74/S3Net-Landslide> (release v6.0.4). CAS imagery remains at <https://doi.org/10.5281/zenodo.10294997>; source terms bind.

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